Contents

1 Markov chains **1**
- 1.1 Definition and basic properties . . . 1
- 1.2 Some technical details . . . 4

2 Communicating classes **5**

3 Hitting times **6**

4 Recurrence and transience **9**
- 4.1 Basic properties . . . 9
- 4.2 Polya's theorem . . . 11

5 Long run behaviour of Markov chains **12**
- 5.1 Invariant distribution . . . 12
- 5.2 Time reversal . . . 15
- 5.3 Convergence to equilibrium . . . 17
- 5.4 The ergodic theorem . . . 19

1 Markov chains

1.1 Definition and basic properties

Definition. A Markov chain is a sequence of random variables $(X_n)_{n=1}^{\infty}$ where the distribution of X_n depends only on X_{n-1}. In other words,

$$P\left(X_{n+1} = x_{n+1} | X_n = x_n, X_{n-1} = x_{n-1}, ..., X_1 = x_1\right) = P\left(X_{n+1} = x_{n+1} | X_n = x_n\right)$$

In general, a sequence of random variables is called a random process or stochastic process.

A Markov chain is called homogeneous if it is also independent of n. In this course, all markov chains will be homogeneous. The state space is the set I of possibilities for X_i. In this course we will always assume the state space is countable to avoid dealing with measure theory weirdness.

Definition. The translation matrix, P, is the matrix such that P_{xy} is the probability of transitioning from x to y.

We say P is a stochastic matrix because each row is a probability distribution because their entries add up to 1.

Example. Suppose the state space is $I = \{0, 1\}$. Then if the matrix is $\begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ then from 0 I go to 1 with probability a and from 1 I go to 0 with probability b.

Example. Suppose the chain is like the graph in Figure 1.

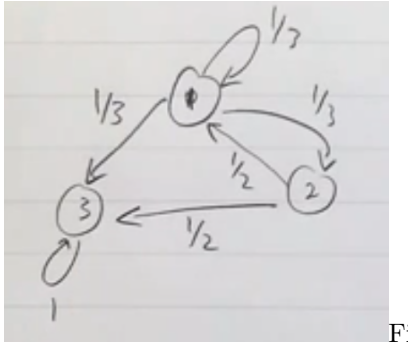


Figure 1

Then the matrix is given by

$$\begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 1 \end{pmatrix}$$

Definition. A process $X = (X_n)$ with values in a discrete set I is said to be $\sim$ Markov (λ, P) if $\lambda = \{\lambda_i : i \in I\}$ is a probability distribution on I and P is a transition matrix and X is a markov chain with transition matrix P and X_0 distributed with $P(X_0 = i) = \lambda_i$.

Proposition. Let I be countable and let λ be a probability distribution on I and let $P = (p_{ij})$ be a stochastic matrix. Then the following are equivalent:

1. X_0 has distribution λ and for every $n \geq 0$ and $x_0, x_1, ..., x_{n+1} \in I$ such that $P(X_0 = x_0, ..., X_n = x_n) > 0$ we have

$$P(X_{n+1} = x_{n+1} | X_0 = x_0, ..., X_n = x_n) = P_{x_n x_{n+1}}$$

This is the precise meaning of "$X \sim$ Markov (λ, P)"

2. For every $n \geq 0$ and $x_0, x_1, ..., x_n \in S$ we have

$$P(X_0 = x_0, ..., X_n = x_n) = \lambda_{x_0} P_{x_0 x_1} ... P_{x_{n-1} x_n}$$

Proof. If we have a Markov chain then whenever $P(X_0 = x_0, ..., X_n = x_n) > 0$ we have

$$P(X_0 = x_0, ..., X_n = x_n) = P(X_n = x_n | X_0 = x_0, ..., X_{n-1} = x_{n-1}) P(X_0 = x_0, ..., X_{n-1} = x_{n-1})$$

$$= P_{x_{n-1} x_n} P(X_0 = x_0, ..., X_{n-1} = x_{n-1})$$

Iterating this gives

$$P(X_0 = x_0, X_1 = x_1, ..., X_n = x_n) = \lambda_{x_0} P_{x_0 x_1} P_{x_1 x_2} ... P_{x_{n-1} x_n}$$

If $P(X_0 = x_0, ..., X_n = x_n) = 0$ then the formula in condition 2 reads "$0 = 0$" so it still holds. To see this, do induction on n. If $P(X_0 = x_0, X_1 = x_1, ..., X_{n-1} = x_{n-1}) = 0$ then (2) holds by the induction hypothesis, and the $n = 0$ case holds by definition. If $P(X_0 = x_0, X_1 = x_1, ..., X_{n-1} = x_{n-1}) > 0$ then we can condition normally and it follows that $P_{x_{n-1} x_n} = 0$ so the product is still 0.

On the other hand, if this formula holds, we have, for any $P(X_n = x_n, ..., X_0 = x_0) > 0$,

$$P(X_{n+1} = x_{n+1} | X_n = x_n, ..., X_0 = x_0) = \frac{P(X_{n+1} = x_{n+1}, X_n = x_n, ..., X_0 = x_0)}{P(X_n = x_n, ..., X_0 = x_0)}$$

$$= \frac{\lambda_{x_0} P_{x_0 x_1} P_{x_1 x_2} ... P_{x_{n-1} x_n} P_{x_n x_{n+1}}}{\lambda_{x_0} P_{x_0 x_1} P_{x_1 x_2} ... P_{x_{n-1} x_n}} = P_{x_n x_{n+1}}$$

□

Example. If we have a sequence of independent random variables[1], then this is a Markov chain, as independence implies independence of the history.

Example. A random walk on $\mathbb{Z}$ is a markov chain.

Definition. For any $i \in I$ we let δ_i denote the probability distribution that puts all the probability mass on i.

Recall that the random variables $X_0, X_1, ..., X_n$ are independent if

$$P(X_0 = x_0, X_1 = x_1, ..., X_n = x_n) = \prod_{i=0}^{n} P(X_i = x_i)$$

[1] See probability and measure for the rigorous proof that this exists in the sense of probability spaces, but here we assume it exists because it is common sense

The processes X_n and Y_n are said to be independent if for every sequence $0 \leq t_1 < t_2 < ... < t_m$ and $0 \leq s_1 < s_2 < ... < s_k$ we have

$$P\left(X_{t_1} = x_1, X_{t_2} = x_2, ..., X_{t_m} = x_m, Y_{s_1} = y_1, ..., Y_{s_k} = y_k\right) = P\left(X_{t_1} = x_1, X_{t_2} = x_2, ..., X_{t_m} = x_m\right) P\left(Y_{s_1} = y_1, ..., Y_{s_k} = y_k\right)$$

for all x_i, y_i.

Proposition. (Extended markov property) Suppose that X_n has Markov (λ, p) and that $m \geq 1$ and $i \in I$. Then conditional on $\{X_m = i\}$ and any history H (provided this history has non-zero probability), the new process $(X_{m+n})_{n \geq 0} \sim$ Markov (δ_i, P) and is independent of $(X_0, X_1, ..., X_m)$. Specifically, for any F given in terms of $\{X_i : i > m\}$ we have

$$P\left(F | X_m = i, H\right) = P\left(F | X_m = i\right)$$

Proof. We have

$$P\left(F | X_m = i\right) = \frac{P\left(X_m = i, F\right)}{P\left(X_m = i\right)}$$

To finish the proof, you can apply the previous proposition. Details are left as an exercise, since it's intuitive from the nature of what a Markov chain actually is.

□

Now assume the state space is finite. We will try to work out $P\left(X_n = x\right)$ in a Markov chain. Naively we can sum over all paths to x.

$$\begin{aligned} P\left(X_n = x\right) &= \sum_{x_0, x_1, ..., x_{n-1}} P\left(X_0 = x_0, X_1 = x_1, ..., X_{n-1} = x_{n-1}, X_n = x_n\right) \\ &= \sum_{x_0, x_1, ..., x_{n-1}} \lambda_{x_0} P_{x_0 x_1} P_{x_1 x_2} ... P_{x_{n-2} x_{n-1}} P_{x_{n-1} x_n} \end{aligned}$$

But notice that this is just the n'th entry of the row vector λP^n. This simplifies the problem a lot.

Note that in a random walk, although the state space is not finite, to find the value at the n'th step we may restrict to things within n of the starting point because we will never reach any other points in the first n steps.

Theorem. If $X \sim$ Markov (λ, P) then $P\left(X_m = x_n\right) = \left(\lambda P^m\right)_n$ and if $P\left(X_m = n\right) > 0$ then $P\left(X_{m+n} = y | X_m = x\right) = \left(P^n\right)_{xy}$

Proof. We already proved the first condition. To prove the second condition, we use the Markov property:

$$P\left(X_{m+n} = y | X_m = x\right) = \left(\delta_x P^n\right)_y = \left(P^n\right)_{xy}$$

□

From now on we will write $P\left(X_{m+n} = y | X_m = x\right) = P\left(X_n = y | X_0 = x\right)$ as $P_{xy}\left(n\right)$.

Example. Suppose the state space is $I = \{0, 1\}$. Then if the matrix is $\begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$ then from 0 I go to 1 with probability a and from 1 I go to 0 with probability b. Suppose we start at 0 and we want to know $P_{00}\left(n\right)$ i.e. the probability of being back at 0 after n steps. Suppose that $a, b > 0$.

Since $P^{n+1} = P_n * P$ we have that $P_{00}\left(n+1\right) = P_{00}\left(n\right)\left(1-a\right) + P_{01}\left(n\right) b$ by considering the possibilities after the first n steps. But then this is $P_{00}\left(n+1\right) = P_{00}\left(n\right)\left(1-a\right) + \left(1 - P_{00}\left(n\right)\right) b$. This is now a recurrence relation, and using $P_{00}\left(0\right) = 1$ we get that $P_{00}\left(n\right) = \frac{b}{a+b} + \frac{a}{a+b}\left(1-a-b\right)^n$. This means that as the number of steps grows, the long term distribution is Bernoulli with probability $\frac{b}{a+b}$.

Now suppose the state space has k elements and that P has k distinct real eigenvalues, so we can write $P = UDU^{-1}$ for D a diagonal matrix. Then we can easily compute $P^n = UD^nU^{-1}$.

Example. Suppose we have a Markov chain on $\{1,2,3\}$ with

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{pmatrix}$$

i.e. from 1 we always go to 2 etc

It turns out that the eigenvalues of P are 1 and $\pm\frac{i}{2}$. Therefore $P^n = U^{-1}\begin{pmatrix} 1 & 0 & 0 \\ 0 & \left(\frac{i}{2}\right)^n & 0 \\ 0 & 0 & \left(-\frac{i}{2}\right)^n \end{pmatrix} U$ for some invertible U. Therefore $p_{11}(n) = A + B\left(\frac{i}{2}\right)^n + C\left(-\frac{i}{2}\right)^n$ for some constants A, B, C. Since it is obviously going to be real-valued, we have that $p_{11}(n) = A + \left(\frac{1}{2}\right)^n \left(B\cos\left(\frac{n\pi}{2}\right) + C\sin\left(\frac{n\pi}{2}\right)\right)$ for some real constants A, B, C that we can solve for. Using $p_{11}(0) = 1$ and $p_{11}(1) = 0$ and $p_{11}(2) = 0$ we get

$$p_{11}(n) = \frac{1}{5} + \left(\frac{1}{2}\right)^n \left[\frac{4}{5}\cos\left(\frac{n\pi}{2}\right) - \frac{2}{5}\sin\left(\frac{n\pi}{2}\right)\right]$$

This tells us that in the long run, $p_{11}(n) \to \frac{1}{5}$.

Write $P_x(\cdot) = P(\cdot|X_0 = x)$ and $E(\cdot) = E(\cdot|X_0 = x)$.

Note that for this eigenvalue method to work the matrix needs to be diagonalizable. Recall from linear algebra that this always happens if the eigenvalues are not repeated or the matrix is symmetric.

1.2 Some technical details

Start with the following theorem which is kind of obvious from common sense but if you want the proper proof we will do it in level 11.1 (probability and measure) as the technical details are surprisingly difficult..

Theorem. Given any sequence F_n of cumulative distribution functions there exists a probability space such that there is a sequence X_n of independent random variables with cdf F_n

Proof. Deferred to probability and measure.

□

We will now prove from this the Kolomogorov extension theorem which justifies later assertions we will make. The main fact to recall from the measure theory part of level 6.2 is uniqueness of finite measures agreeing on a generating π-system.

Theorem. (Kolmogorov extension theorem, countable state version) Let I be a non-empty countable set. For each $n \geq 0$ let p_n be a probability mass function on I^{n+1}. Suppose that these are consistent in the sense that

$$\sum_{j \in I} p_{n+1}(x_0, ..., x_n, j) = p_n(x_0, ..., x_n)$$

Then there exists a unique probability measure on $I^{\mathbb{Z}_{\geq 0}}$ with its coordinate-generated sigma-algebra with the property that

$$Q(w_0 = x_0, ..., w_n = x_n) = p_n(x_0, ..., x_n)$$

Proof. Take independent uniform $(0,1)$ random variables $U_0, U_1, ...$ by the theorem above. We can sample any distribution on I using a uniform variable by enumerating I and dividing $(0,1)$ into intervals of the prescribed probabilities.

Fix $i_* \in I$. For each finite history $a = (x_0, ..., x_n)$ define the probability distribution

$$q_n(a, j) = \begin{cases} \frac{p_{n+1}(a,j)}{p_n(a)} & p_n(a) > 0 \\ 1_{\{j = i_*\}} & p_n(a) = 0 \end{cases}$$

By consistency, we have $\sum_j q_n(a, j) = 1$ universally.

Use U_0 to sample X_0 with distribution p_0. Recursively use U_{n+1} to sample X_{n+1} from $q_n((X_0, ..., X_n), \cdot)$. These variables are measure because the interval sampling maps are measurable and there are only countably many histories.

Inductively, independence gives

$$P((X_0, ..., X_n) = a, X_{n+1} = j) = P((X_0, ..., X_n) = a) q_n(a, j)$$

$$= p_n(a) q_n(a, j) = p_{n+1}(a, j)$$

This also holds when $p_n(a) = 0$ because consistency then forces $p_{n+1}(a, j) = 0$.

Now define

$$Q(B) = P((X_0, X_1, ...) \in B)$$

This makes sense because we have a sequence of uniform random variables and is countably additive because pre-images preserve disjoint countable unions and P is countably additive. It has the required finite-dimensional distributions, and uniqueness follows from the generating π-system result, applied to finite-history cylinders together with the empty set.

□

2 Communicating classes

Definition. We say x leads to y or $x \to y$ if $P_x(x_n = y \text{ for some } n \geq 0) > 0$.

Definition. We say x communicates with y or $x \longleftrightarrow y$ if x leads to y and y leads to x.

Proposition. The following are equivalent

1. x leads to y

2. There exists a sequence of states $x = x_0, x_1, ..., x_n = y$ such that all transition steps have positive probability

3. $P_n(X_n = y) > 0$ for some $n \geq 0$

Proof. (1) $\iff$ (3): The event $\{X_n = y \text{ for some } n \geq 0\}$ can be expressed as $\cup_{n \geq 0} \{X_n = y\}$ so if $\{X_n = y \text{ for some } n \geq 0\} > 0$ there exists some $n \geq 0$ such that $P_x(X_n = y) > 0$. It is obvious that if this is true for some n then $x \to y$.

(2) $\iff$ (3): Since

$$p_{xy}(n) = \sum_{x_0, x_1, ..., x_{n-1}} \lambda_x P_{xx_1} P_{x_1 x_2} ... P_{x_{n-2} x_{n-1}} P_{x_{n-1} y}$$

this is non-zero if and only if there is a path such that all $P_{x_i x_{i+1}}$ is non-zero.

□

Proposition. The relation $\longleftrightarrow$ is an equivalence relation.

Call the classes communicating classes.

Proof. Reflexivity and symmetry follow by definition, reflexivity from condition (3) of the previous proposition. By condition (2) of the previous proposition, transitivity follows.

□

Example. Suppose the state space is $\{1,2,3,4,5\}$ and the transition matrix is

$$\begin{pmatrix} & & 0.5 & 0.5 & \\ 0.5 & 0.5 & & & \\ & 0.5 & 0.5 & & \\ & & & 0.5 & 0.5 \\ & & & 1 & \end{pmatrix}$$

Then the communicating classes are $\{1,2,3\}$ and $\{4,5\}$. Notice that we can get from $\{1,2,3\}$ to $\{4,5\}$ but cannot come back, so we can do whatever in $\{1,2,3\}$ but in the long term we will be stuck in $\{4,5\}$.

Definition. A communicating class C is said to be closed if $(x, \in C, x \to y) \implies (y \in C)$

Definition. A Markov chain is said to be irreducible if there is only 1 communicating class.

Definition. A state $x \in I$ is absorbing if $\{x\}$ is a closed class i.e. $p_{xx}(n) = 1$ for all n

3 Hitting times

Definition. Let $A \subseteq I$. Then the first hitting time T_A is $\inf\{n \geq 0 : X_n = A\}$ (this is ∞ if A is never visited). The hitting probabilities $h^A : I \to [0,1]$ are $h_i^A = P_i(T_A < \infty)$. The mean hitting time is $K_i^A : I \to [0,\infty]$ is $E_i(T_A)$.

Theorem. Suppose X is a Markov chain on a state space I with transition matrix P. If $A \subseteq I$ then the hitting probabilities h_i^A of gettting to A starting at i (assume probability of $X_0 = i$ is > 0) satisfies

$$h_i^A = \begin{cases} 1 & i \in A \\ \sum_{j \in I} P(i,j) h_j^A & i \notin A \end{cases}$$

Moreover, h_i^A is the minimal non-negative solution to this system of equations in the sense that if x_i is another non-negative solution then $h_i^A \leq x_i$ for all $i \in I$.

Proof. The fact that h_i^A is 1 if $i \in A$ is obvious so suppose $i \notin A$. Define $J_i = \{j \in I : P(i,j) > 0\}$, then

$$h_i^A = P_i\left(H^A < \infty\right) = \sum_{j \in J_i} P_i\left(H^A < \infty | X_1 = j\right) p_{i,j} = \sum_{j \in J_i} h_j^A p_{i,j} = \sum_{j \in I} h_j^A p_{i,j}$$

so h_i^A is indeed a solution to the equations. We have to show that it is minimal.

Let x be a non-negative solution *i.e.*

$$x_i = \begin{cases} 1 & i \in A \\ \sum_{j \in I} P(i,j) x_j & i \notin A \end{cases}$$

If $i \in A$ we have $h_i^A = x_i = 1$. Otherwise, we can write

$$x_i = \sum_j p_{i,j} x_j = \sum_{j \in A} p_{i,j} x_j + \sum_{j \notin A} p_{i,j} x_j = \sum_{j \in A} p_{i,j} + \sum_{j \notin A} p_{i,j} x_j \geq \sum_{j \in A} p_{i,j} = P_i\left(H^A = 1\right)$$

This is allowed because if $p_{i,j} = 0$ the terms will contribute nothing anyway.

By iterating this process, we get

$$x_i = \sum_{j \in A} p_{i,j} + \sum_{j \notin A} p_{i,j} \left(\sum_k p_{i,k} x_k\right) = \sum_{j \in A} p_{i,j} + \sum_{j \notin A} p_{i,j} \left(\sum_{k \in A} p_{i,k} x_k + \sum_{k \notin A} p_{i,k} x_k\right)$$

$$\geq P_i\left(H^A = 1\right) + \sum_{j \notin A, k \in A} p_{i,j} p_{j,k} = P_i\left(H^A = 1\right) + P_i\left(H^A = 2\right)$$

By induction we obtain

$$x_i \geq P_i\left(H^A \leq n\right)$$

for all n. Taking the limit as $n \to \infty$ we get

$$x_i \geq P_i\left(H^A < \infty\right) = h_i^A$$

so h_i^A is minimal.

□

Example. Suppose we have a markov chain with 4 states $0, 1, 2, 3$ and if we are at 0 or 3 we stay there and otherwise we move to the adjacent bumber with probability $\frac{1}{2}$.

Let A be the closed communicating class 3. We want to find the probability of getting to 3 from 1 and from 2. Write

$$P = \begin{pmatrix} 1 & & & \\ \frac{1}{2} & & \frac{1}{2} & \\ & \frac{1}{2} & & \frac{1}{2} \\ & & & 1 \end{pmatrix}$$

By the above theorem we can write down the system of equations:

First, by common sense, $h_0(A) = 0$ and $h_3(A) = 1$. Then we have

$$h_1(A) = \frac{1}{2}\left(h_0(A) + h_2(A)\right) = \frac{1}{2}h_2(A)$$

$$h_2(A) = \frac{1}{2}\left(h_1(A) + h_3(A)\right) = \frac{1}{2} + \frac{1}{2}h_1(A)$$

solving this gives $h_1(A) = \frac{1}{3}$ and $h_2(A) = \frac{2}{3}$.

Example. Consider the random walk on $\mathbb{Z}_{\geq 0}$ where $p(i, i+1) = p$ and $p(i, i-1) = 1-p$ and $p(0, 1) = 1$. Then we want to ask $h_i^{\{0\}} =: h_i$.

From the theorem, we have

$$h_i = ph_{i+1} + (1-p)h_{i-1}$$

for $i \geq 1$ and $h_0 = 1$. If $p \neq 1-p$ then the general solution of this is

$$h_i = a + b\left(\frac{1-p}{p}\right)^i$$

If $p > \frac{1}{2}$ then this means by minimality and non-negativity and the initial condition we must have $a = 0$ and $b = 1$ so that

$$h_i = \left(\frac{1-p}{p}\right)^i$$

so this goes to 0 as we go further to the right, as our intuition suggests.

On the other hand, if $p < \frac{1}{2}$ then minimality and non-negativity forces us to take $a = 1$ and $b = 0$ which means we will hit 0 with probability 1.

If $p = \frac{1}{2}$ the genera solution becomes of the form $h_i = a + bi$ and minimality + non-negativity + the initial condition implies $a = 1$ so $h_i^A = 1$ for all $i \geq 1$.

Definition. k_i^A is $E(T_A)$ and T_A is the first hitting time of A.

Example. Suppose X is a markov chain on $I = \{0, 1, 2, ...\}$ with transition probabilities given by p_i to go from i to $i+1$ and a probability $1 - p_i$ to go from i to $i-1$. The difference this time is that the probability depends on i. If we get to 0 in X, we will stay there with probability 1. We will investigate the probability of reaching 0. Suppose all other p_i's are not 0.

We know that $h_i^A = p_i h_{i+1}^A + (1-p_i) h_{i-1}^A$. Write this as

$$p_i \left(h_{i+1}^A - h_i^A\right) = (1-p_i)\left(h_i^A - h_{i-1}^A\right)$$

This is a recursion for $u_i := h_{i+1}^A - h_i^A$. Write $\gamma_i := \prod_{j=1}^{i} \frac{1-p_j}{p_j}$, then we have $u_{i+1} = \gamma_i u_1$. This implies that $h_i^A = 1 + \sum_{j=1}^{i} u_j = \left(\sum_{j=1}^{i-1} \gamma_i\right) u_1 + 1$. We have $u_1 = h_1 - 1$ which implies

$$h_i^A = 1 + \sum_{j=1}^{i} u_j = 1 - \left(\sum_{j=1}^{i-1} \gamma_i\right)(1-h_1)$$

Since h must be non-negative, we must have

$$(1-h_1) \leq \frac{1}{\sum_{j=0}^{\infty} \gamma_j}$$

To achieve minimality we therefore need to take

$$h_1 = 1 - \frac{1}{\sum_{j=0}^{\infty} \gamma_j}$$

Therefore, if $\sum_{j=0}^{\infty} \gamma_j$ does not converge, then $h_i = 1$ for all $i \geq 1$ but if it converges then $h_i = \frac{\sum_{j=i}^{\infty} \gamma_j}{\sum_{j=0}^{\infty} \gamma_j}$

Theorem. Suppose X is a markov chain with transition matrix P and let $A \subseteq I$. Then the mean hitting time of A from i, denoted K_i^A, is the minimal non-negative solution to

$$K_i^A = \begin{cases} 0 & i \in A \\ 1 + \sum_{j \notin A} P(i,j) K_j^A & i \notin A \end{cases}$$

Assume $P(X_0 = i) \neq 0$.

Proof. Assume $i \notin A$ since for $i \in A$ it is clearly 0.

Using the formula $E[X] = \sum P(X > n)$, we have

$$K_i^A = \sum_{n=0}^{\infty} P_i(T_A > n) = \sum_{n=0}^{\infty} P(X_0 \notin A, X_1 \notin A, ..., X_n \notin A | X_0 = i)$$

$$= 1 + \sum_{n=1}^{\infty} P(X_0 \notin A, X_1 \notin A, ..., X_n \notin A | X_0 = i)$$

$$= 1 + \sum_{n=1}^{\infty} \sum_{j:P(i,j)>0} P(i,j) P(X_0 \notin A, X_1 \notin A, ..., X_n \notin A | X_0 = i, X_1 = j)$$

$$= 1 + \sum_{n=1}^{\infty} \sum_{j:P(i,j)>0} P(i,j) P_j(X_0 \notin A, X_1 \notin A, ..., X_{n-1} \notin A)$$

$$= 1 + \sum_{n=1}^{\infty} \sum_{j:P(i,j)>0} P(i,j) P_j(T_A > n-1) = 1 + \sum_{j:P(i,j,)>0} P(i,j) K_j^A$$

$$= 1 + \sum_{j \notin A} P(i,j) K_j^A$$

the last equality is because adding in terms where $P(i,j) = 0$ does nothing and is fine as we are no longer conditioning, and removing terms where $j \in A$ does nothing as K_j^A is 0 for all such j.

Next, we prove minimality. Let x_i be another non-negative solution to the system of linear equations. Then $x_i = 0$ for $i \in A$ and iterating the linear equations we get

$$x_i = 1 + \sum_{j_1 \notin A} P(i, j_1) + \sum_{j_1 \notin A} \sum_{j_2 \notin A} P(i,j_1) P(j_1, j_2) + ... + \sum_{j_1, j_2, ..., j_n \notin A} P(i,j_1) P(j_1,j_2) ... P(j_{n-1}, j_n)$$

$$+ \sum_{j_1, j_2, \ldots, j_n, j_{n+1} \notin A} P(i, j_1) P(j_1, j_2) \ldots P(j_{n-1}, j_n) x_{j_{n+1}}$$

Since x_j is non-negative for all j, we get

$$x_i \geq 1 + \sum_{j_1 \notin A} P(i, j_1) + \sum_{j_1 \notin A} \sum_{j_2 \notin A} P(i, j_1) P(j_1, j_2) + \ldots + \sum_{j_1, j_2, \ldots, j_n \notin A} P(i, j_1) P(j_1, j_2) \ldots P(j_{n-1}, j_n)$$

$$= 1 + \sum_{r=1}^{n} P_i(T_A > r) = \sum_{r=0}^{n} P_i(T_A > r)$$

By taking the limit in n, we get

$$x_i \geq \sum_{r=0}^{\infty} P_i(T_A > r)$$

This implies $x_i \geq K_i^A$.

□

Definition. A random variable T with values in $0 \cup \mathbb{N} \cup \infty$ is a stopping time with respect to a random provess $X = (X_n)_{n \geq 0}$ if the event $\{T = n\}$ only depends on $(X_0, \ldots, X_n)$.

Example. T_A, the hitting time, is a stopping time.

Theorem. (Strong Markov property) Let X be Markov (λ, P) and let T be a stopping time. Then, conditional on $T < \infty$ and $X_T = i$ (assuming the probability of this is > 0), the process X_{T+n} is Markov (δ_i, P) and it is independent of $X_0, \ldots, X_T$.

Proof. Lowkey obvious from common sense, just remember T has to be a stopping time so it cannot involve using information about the future by saying something like "T is the last time i is hit". [2]

□

Now consider the simple symmetric random walk on $\mathbb{Z}$, with the version where we bounce back to 1 if we are at 0. We know from earlier that the probability of reaching 0 if we start at 0 is 1, but what if we start at 2? Well, we know that the probability of getting to 0 from 1 is 1, since the probability of going back to 0 from 0 is also 1. But we know that the probability of going to 1 from 2 is also 1, as this is essentially the same problem by the strong markov property, and then the problem of getting to 0 again has probability 1.

4 Recurrence and transience

4.1 Basic properties

A state $i \in I$ is recurrent if the probability that $X_n = i$ for infinitely many i is 1, and a state is transient if it is not recurrent.

We define $T_i^{(0)} = 0$ and for $k \geq 0$ $T_i^{(k+1)}$ to be the first time we visit i after $T_i^{(k)}$. i.e. the time we get to i the $k+1$'th time. The return probability is $f_i := P_i\left(T_i^{(1)} < \infty\right)$, and V_i is the total number of visits, including the starting visit.

Lemma. For every $r \geq 0$ we have $P_i(V_i > r) = f_i^r$.

Proof. By induction: This is obvious for $r = 0$ by definition. If true for r then

$$P_i(V_i > r+1) = P_i(v_i > r+1, v_i > r) = P_i(v_i > r+1 | v_i > r) P_i(v_i > r)$$

[2] If studying for the Cambridge course, this proof isn't suitable for exams - see someone else's notes for the exam proof. As this whole website is an exposition of why maths is true and not an exam guide, we will stick with this proof.

so done by strong Markov property. If $P_i(v_i > r)$ is 0 then just argue that $P_i(V_i > r+1)$ is a sub-event.

□

Proposition. The probability that $X_n = i$ for infinitely many i is either 1 or 0

Proof. Follows from the previous lemma: If $f_i < 1$ then $P_i(V_i = \infty) = \lim f_i^r = 0$, and if $f_i = 1$ then the probability that $V_i = \infty$ is 1 by countable additivity.

□

Theorem. For a markov chain X on I, a state i is recurrent if and only if $\sum_{n\geq 0} P_{ii}(n) = \infty$

Proof. Note that

$$E_i(V_i) = E_i\left(\sum_{k=0}^{\infty} 1(X_k = i)\right) = \sum_{k=0}^{\infty} P(X_k = i) = \sum_{k=0}^{\infty} P_{ii}(k)$$

If this expectation is finite then the probability that V_i is infinite is 0 so we have transience. On the other hand if the expectation is infinite we have recurrence as V_i is known to follow a geometric distribution.

□

Theorem. If x communicates with y then they are either both transient or both recurrent. ("recurrence is a class property")

Proof. Suppose x is recurrent. Since $x \longleftrightarrow y$ there are integers k, l such that $P_{xy}(k) > 0$ and $P_{yx}(l) > 0$. Then

$$\sum_{n\geq 0} P_{yy}(n) \geq \sum_{n\geq 0} P_{yy}(l+n+k) \geq \sum_{n\geq 0} P_{yx}(l) P_{xx}(n) P_{xy}(k) = P_{yx}(l) P_{xy}(k) \sum_{n\geq 0} P_{xx}(n)$$

Since x is recurrent this sum is infinite and thus so is the y sum. So the states are either both or neither recurrent.

□

Example. This means that in a random walk we will visit every number infinitely often with probability 1.

Proposition. A recurrent communicating class is closed.

Proof. Suppose not i.e. suppose that $x \in C$ is recurrent and $x \to y$ and $y \notin C$. Then we can never get back or else x and y would communicate. Then there is m with $P_{xy}(m) > 0$ so $P_x(V_x < \infty) \geq P(x_m = y) < 0$ so not closed implies transient.

□

Proposition. Suppose $C \subseteq I$ is a closed communicating class. Then if C is finite, then C is recurrent.

Proof. Fix $x \in C$. Then since C is closed, there must be a $y \in C$ such that $P_x(X_n = y \text{ infinitely often}) > 0$. Since C is a communicating class, $x \longleftrightarrow y$ so there is $m \geq 1$ such that $P_{yx}(m) > 0$. Then

$$P_y(X_n = y \text{ i.o.}) \geq P_y(X_m = x) P_x(X_{n+m} = y \text{ i.o. }) > 0$$

Therefore y is recurrent and thus so is everything because recurrence is a class property.

□

Proposition. If $A \subseteq I$ is a recurrent class then $P_x(T_y < \infty) = 1$ for all $x, y \in A$.

Proof. Choose x, y and pick $m \geq 1$ such that $P_{yx}(m) > 0$. Then by recurrence we have

$$1 = P_y(X_n = y \text{ i.o.}) = \sum_{z \in I} P_y(x_m = z, X_n = y \text{ for infinitely many } n \geq m) = \sum_{z \in I} P_{yz}(m) P_z(X_n = y \text{ for infinitely many } n)$$

By the strong markov property at time T_y we get

$$= \sum_{z \in I} P_{yz}(m) P_z(T_y < \infty) P_y(X_n = y \text{ for infinitely many } n) = \sum_{z \in I} P_{yz}(m) P_z(T_y < \infty)$$

Since $\sum_{z \in I} P_{yz}(m) = 1$ and $P_{yx}(m) > 0$ this forces $P_x(T_y < \infty) = 1$.

□

4.2 Polya's theorem

Definition. A simple symmetric random walk on $\mathbb{Z}^d$ is the Markov chain with transition probabilities $\frac{1}{2d}$ to go to each neighbour, starting at 0.

Note that since a random walk is the only communicating class, every point will be recurrent or transient.

Definition. If $\sum_i a_i = n$ then $\begin{pmatrix} n \\ a_1, a_2, ..., a_n \end{pmatrix} = \frac{n!}{a_1! a_2! ... a_n!}$

Theorem. (Polya's theorem) The simple symmetric random walk is recurrent for $d = 1, 2$ and transient for $d = 3$.

Proof. We will do $d = 2$ since it clearly implies the $d = 1$ case. For thie case, note that the "sum of coordinates" and "difference of coordinates" are both simple random walks on $\mathbb{Z}$ and they are independent. Therefore it can be thought of as 2 independent simple random walks on $\mathbb{Z}$.

For a $1D$ random walk to get home there are $\begin{pmatrix} 2n \\ n \end{pmatrix}$ ways after $2n$ steps so $\sum_n p_{00}(2n) = \sum \frac{(2n)!}{(n!)^2} \frac{1}{2^{2n}}$. By Stirling's approximation this is $\sim \frac{1}{\sqrt{n}}$, so the probability that it happens for 2 random walks simultaneously is $\sim \frac{1}{n}$ and these sums both diverge.

On the other hand, we want to show a $3D$ random walk is transient. Note that X can be 0 only at even times. We have

$$p_{00}(2n) = \sum_{i,j,k \geq 0, i+j+k=n} \begin{pmatrix} 2n \\ i,i,j,j,k,k \end{pmatrix} \left(\frac{1}{6}\right)^{2n} = \begin{pmatrix} 2n \\ n \end{pmatrix} \left(\frac{1}{2}\right)^{2n} \sum_{i,j,k \geq 0, i+j+k=n} \begin{pmatrix} n \\ i,j,k \end{pmatrix}^2 \left(\frac{1}{3}\right)^{2n}$$

Therefore we have that

$$\sum_{i,j,k \geq 0, i+j+k=n} \begin{pmatrix} n \\ i,j,k \end{pmatrix} \left(\frac{1}{3}\right)^n = 1$$

since the sum corresponds to the total probability of all ways of placing n balls into 3 boxes uniformly at random. If $n = 3m$ it is an exercise to show that

$$\begin{pmatrix} n \\ i,j,k \end{pmatrix} \leq \begin{pmatrix} n \\ m,m,m \end{pmatrix}$$

Therefore we have by stirling that

$$p_{00}(2n) \leq \begin{pmatrix} 2n \\ n \end{pmatrix} \left(\frac{1}{2}\right)^{2n} \begin{pmatrix} n \\ m,m,m \end{pmatrix} \left(\frac{1}{3}\right)^n \sim \frac{C}{n^{3/2}}$$

Therefore we have that the sum of $p_{00}(6m)$ is finite and using that $p_{00}(6m) \sim p_{00}(6m \pm 2)$ we get that the whole sum is finite which implies the result.

□

5 Long run behaviour of Markov chains

5.1 Invariant distribution

We want to understand how $P_x(X_n = y)$ behaves for large n.

If $P(X_n = j) \to \pi_j$ as $n \to \infty$ for all $j \in I$ then we expect that if $X_n \sim \pi$ then X_{n+1} should also have distribution π i.e.

$$\pi_j = P(X_{n+1} = j) = \sum_i P(X_n = i, X_{n+1} = j) = \sum \pi_i p_{ij}$$

Motivated by this, we define

Definition. π is an invariant distribution if $\pi = \pi P$.

Proposition. If π is invariant for some markov chain X and $X_0 \sim \pi$ then $X_n \sim \pi$ for all n.

Proof. This follows by the discussion above and by induction.

□

Theorem. If X is a Markov chain on a finite state space I and $P(X_n = i)$ converges to some π_i as $n \to \infty$ for all $i \in I$ then π is an invariant distribution for X.

Proof. We have

$$P(X_{n+1} = j) = \sum_{i \in I} P(X_n = i, X_{n+1} = j) = \sum_{i \in I} P(X_n = i) P_{ij}$$

By assumption, $P(X_{n+1} = j) \to \pi_j$ and $P(X_n = i) \to \pi_i$. Since I is finite, taking limits in this equation gives that $\pi_j = \sum_{i \in I} \pi_i p_{ij}$ so π is invariant. Also, because I is finite, considering limits implies that π actually is a probability distribution in the sense $\sum \pi_i = 1$. Note that finiteness is essential because on a symmetric random walk, π_i would be 0 for all i so their sum isn't 1.

□

Lemma. (Brouwer fixed point theorem) Any continuous function from a non-empty, compact, convex set to itself has a fixed point

This is actually analysis, but it will be used in the proof of the next theorem.

Proof. Note: We will see how to prove this formally in Algebraic Topology, but for now here is why it is true allowing for common sense.

Let f be such a map with no fixed point. Suppose we have a point x in the set, then the map f will map it somewhere. If this is a fixed point then done, otherwise there is a straight line between them. Let $g(x)$ be the unique point where this line intersects the boundary of the set but closer to x than $f(x)$. If there is no fixed point, this is well defined and continuous and it is a map that fixes the boundary of the convex set and mapping every point onto the boundary, which is clearly nonsense because you would have to tear the middle to do this which results in the map being discontinuous.

□

Theorem. Every Markov chain on a finite state space has an invariant distribution.

Proof. Define the set of all probability distributions over a finite state space of size n as the standard simplex Δ^{n-1}:

$$\Delta^{n-1} = \left\{ x \in \mathbb{R}^n \mid x_i \geq 0 \text{ for all } i, \sum_{i=1}^{n} x_i = 1 \right\}$$

This set Δ^{n-1} is compact and convex in $\mathbb{R}^n$.

Define a function $f : \Delta^{n-1} \to \Delta^{n-1}$ by matrix multiplication: $f(x) = xP$

Because P has non-negative entries and maps probability vectors to probability vectors since

$$(xP)_j = \sum_i x_i P_{ij} \geq 0 \text{ and } \sum_j \sum_i x_i P_{ij} = \sum_i x_i \sum_j P_{ij} = \sum_i x_i = 1$$

the map f is well-defined, bounded, and continuous.

By the previous lemma there exists a vector $\pi \in \Delta^{n-1}$ such that: $\pi = f(\pi) = \pi P$. This π is an invariant probability distribution.

□

Definition. For $k, i \in I$ define $\nu_k(i) := E_k\left[\sum_{n=0}^{T_k-1} 1(X_n = i)\right]$. This is the expected number of visits to i before returning to k when starting from k.

Theorem. Suppose that P is irreducible and recurrent. Then ν_k is an invariant measure i.e. $\nu_k = \nu_k P$, satisfying $0 < \nu_k(i) < \infty$ and $\nu_k(k) = 1$.

Proof. By the definition of ν_k it is clear that $\nu_k(k) = 1$.

Since the chain is recurrent, $T_k < \infty$ with probability 1 and since $X_{T_k} = k$ by definition of T_k we obtain

$$\nu_k(i) = E_k\left[\sum_{n=1}^{T_k} 1(X_n = i)\right] = E_k\left[\sum_{n=1}^{\infty} 1(n \leq T_k) 1(X_n = i)\right] = \sum_{n=1}^{\infty} P_k(X_n = i, n \leq T_k)$$

Therefore for all $n \geq 1$ we have

$$P_k(X_n = i, n \leq T_k) = \sum_j P_k(X_n = i, X_{n-1} = j, n \leq T_k)$$

Now note that the event $\{T_k \geq n\}$ depends only on $X_0, X_1, ..., X_{n-1}$. Note that the equality we are establishing in the following line holds trivially just says $0 = 0$ if $P(X_{n-1} = j, n \leq T_k) = 0$, so assuming that is not 0 we have

$$P_k(X_n = i, X_{n-1} = j, n \leq T_k) = P_k(X_n = i | X_{n-1} = j, n \leq T_k) P_k(X_{n-1} = j, n \leq T_k) = P(j,i) P(X_{n-1} = j, n \leq T_k)$$

Therefore putting this all together we have

$$\nu_k(i) = \sum_{n=1}^{\infty} \sum_j P(j,i) P(X_{n-1} = j, n \leq T_k) = \sum_j P(j,i) \sum_{n=1}^{\infty} E_k[1(X_{n-1} = j) 1(n \leq T_k)]$$

$$= \sum_j P(j,i) E_k\left[\sum_{n=0}^{T_k-1} 1(X_n = j)\right] = (\nu_k P)_i$$

This implies ν_k is an invariant measure.

To show that $0 < \nu_k(i) < \infty$ let m, n be such that $p_{ki}(n) > 0$ and $p_{ik}(m) > 0$ which exists by irreducibility. Then using the invariance of ν_k we get

$$\nu_k(i) \geq \nu_k(k) p_{ki}(n) = p_{ki}(n) > 0$$

since $\nu_k(k) = 1$. To prove finiteness, note that by invariance we have $\nu_k P^m = \nu_k$ so

$$1 = \nu_k(k) \geq p_{ik}(m) \nu_k(i) \implies \nu_k(i) \leq \frac{1}{p_{ik}(m)} < \infty$$

□

Theorem. Suppose that P is an irreducible translation matrix and λ is an invariant measure satisfying $\lambda_k = 1$. Then $\lambda \geq \nu_k$. If P is also recurrent them $\lambda = \nu_k$.

Proof. Since λ is invariant we get

$$\lambda_i = P(k,i) + \sum_{j_1 \neq k} \lambda_{j_1} P(j_1,i)$$

$$= P(k,i) + \sum_{j_1 \neq k} P(k,j_1) P(j_1,i) + \sum_{j_1,j_2 \neq k} P(j_2,j_1) P(j_1,i) \lambda_{j_2}$$

$$= P(k,i) + \sum_{j_1 \neq k} P(k,j_1) P(j_1,i) + ... + \sum_{j_1,j_2,...,j_{n-1} \neq k} P(k,j_{n-1}) ... P(j_2,j_1) P(j_1,i)$$

$$+ \sum_{j_1,j_2,...,j_n \neq k} \lambda_{j_n} P(j_n,j_{n-1}) P(j_{n-1},j_{n-2}) ... P(j_2,j_1) P(j_1,i)$$

Since λ is a measure we must have $\lambda_x \geq 0$ for all x and therefore for all $i \neq k$ we have

$$\lambda_i \geq P(k,i) + \sum_{j_1 \neq k} P(k,j_1) P(j_1,i) + ... + \sum_{j_1,j_2,...,j_{n-1} \neq k} P(k,j_{n-1}) ... P(j_2,j_1) P(j_1,i)$$

$$= P_k(X_1 = i, T_k \geq 1) + P_k(X_2 = i, T_k \geq 2) + ... + P_k(X_n = i, T_k \geq n)$$

$$= \sum_{l=1}^{n} P_k(X_l = i, T_k \geq l) \to \sum_{l=1}^{\infty} P_k(X_l = i, T_k \geq l) = \nu_k(i)$$

This proves the first part. Now suppose that P is also recurrent. Then ν_k is an invariant measure by the theorem above so $\lambda - \nu_k$ is also invariant since $\lambda \geq \nu_k$ and it satisfies $\lambda_k - \nu_k(k) = 0$. By irreducibility there exists $m > 0$ such that $p_{ik}(m) > 0$. Using invariance we get

$$0 = \lambda_k - \nu_k(k) = \sum_j (\lambda_j - \nu_k(j)) p_{jk}(m) \geq (\lambda_i - \nu_k(i)) p_{ik}(m)$$

This implies that $\lambda_i = \nu_k(i)$.

□

Given an invariant measure, we can divide by $\sum_{i \in I} \nu_k(i)$ (provided this is finite) to get an invariant distribution. We calculate

$$\sum_{i \in I} \nu_k(i) = \sum_{i \in I} E_k \left[\sum_{n=0}^{T_k - 1} 1(X_n = i) \right] = E_k \left[\sum_{n=0}^{T_k - 1} \sum_{i \in I} 1(X_n = i) \right] = E_k[T_k]$$

Therefore in order to get an invariant distribution this way we need $E_k[T_k]$ to be finite. Motivated this we define

Definition. A state is positive recurrent if the expected first return time is finite and null recurrent otherwise.

Theorem. Suppose that P is an irreducible transition matrix. Then the following are equivalent:

1. Every state is positive recurrent

2. Some state is positive recurrent

3. P has an invariant distribution π.

and when this happens, $\pi_i = \frac{1}{E_i[T_i]}$ for every i. In particular, π_i is always non-negative so if a later proof were to divide by it (this happens) it would be fine.

Proof. $1 \implies 2$: Obvious

$2 \implies 3$: Let k be a positive recurrent state and consider the invariant measure ν_k, then as in the above discussion we can define $\pi_i = \frac{\nu_k(i)}{E_k[T_k]}$ then π is invariant

$3 \implies 1$: Let k be a state, then we want to show that k is positive recurrent. First we show that $\pi_k > 0$. Let $i \in I$ be such that $\pi_i > 0$ and by irreducibility there exists $n \geq 0$ such that $p_{ik}(n) > 0$. By stationarity of π we get

$$\pi_k = \sum_{j \in I} \pi_j p_{jk}(n) \geq \pi_i p_{ik}(n) > 0$$

so we can define a new invariant measure $\lambda_i = \frac{\pi_i}{\pi_k}$ for every $i \in I$. Note that $\lambda_k = 1$ so by the previous theorem we get $\lambda \geq \nu_k$ which implies that

$$E_k[T_k] = \sum_{i \in I} \nu_k(i) \leq \sum_{i \in I} \lambda_i = \frac{1}{\pi_k} < \infty$$

This proves that k is positive recurrent.

For the last part:

Let $k \in I$. Then k is positive recurrent by (i) and therefore also recurrent. This means that the measure λ that we defined must be equal to ν_k by the previous theorem. Thus we obtain that

$$\sum_{i \in I} \frac{\pi_i}{\pi_k} = \sum_{i \in I} \nu_k(i) = E_k[T_k]$$

which, using that π is a distribution, gives

$$\pi_k = \frac{1}{E_k[T_k]}$$

□

Corollary. We always have $\nu_x(y) = \frac{\pi(y)}{\pi(x)}$ i.e. we expect to visit y that many times between each visit of x.

Example. Consider the simple symmetric random walk on $\mathbb{Z}$, then $\pi_i = 1$ for all $i \in \mathbb{Z}$ is an invariant measure. Then since P is a recurrent walk we get that all invariant measures are multiples of π which implies that X is not positive recurrent since $\sum_{i \in \mathbb{Z}} \pi_i = \infty$.

Example. Let X be an asymmetric random walk on $\mathbb{Z}$ with probability p to go forwards and $q = 1 - p$. Then by invariantness we have $\pi_i = \pi_{i-1} p + \pi_{i+1} q$ which we can solve to get $\pi_i = a + b\left(\frac{p}{q}\right)^i$. This means uniqueness up to multiplicative constants does not hold, so this walk is transient.

Example. Let X be a random walk with probability p as above with the difference that we cannot go from 0 to -1 and instead stay at 0 with probability q. Then if we try to solve $\pi = \pi P$ the equations we get are

$$\pi_0 = q\pi_1 + q\pi_0$$

$$\pi_k = p\pi_{k-1} + q\pi_{k+1} : k \geq 1$$

Solving this gives $\pi_1 = \pi_0 \frac{p}{q}$ and $\pi_k = \left(\frac{p}{q}\right)^k \pi_0$. We can normalize π to get a probability distribution by taking $\pi_0 = 1 - \frac{p}{q}$ because then we get

$$\pi_k = \left(\frac{p}{q}\right)^k \left(1 - \frac{p}{q}\right)$$

which implies that the distribution is indeed invariant and that X is positive recurrent.

5.2 Time reversal

Proposition. Let X be a markov chain with transition matrix P which is irreducible and has invariant distribution π. Let $N \in \mathbb{N}$ and suppose that $X_0 \sim \pi$. Then the process $Y_n := X_{N-n}$ for $0 \leq n \leq N$ is a Markov chain with transition matrix $\hat{P}$ given by

$$\hat{P}(x, y) := \frac{\pi(y)}{\pi(x)} P(y, x)$$

Also, $\hat{P}$ is irredicible and has invariant distribution π.

Proof. Note that

$$\sum_y \hat{P}(x,y) = \sum_y \frac{\pi(y)}{\pi(x)} P(y,x) = \frac{\pi(x)}{\pi(x)} = 1$$

since π is invariant, which implies that $\hat{P}$ is the transition matrix of *some* markov chain.

To show that Y is a Markov chain:

$$P(Y_0 = y_0, ..., Y_n = y_n) = P(X_0 = y_N, ..., X_N = y_0)$$

$$= \pi(y_N) P(y_N, y_{N-1}) ... P(y_1, y_0) = \pi(y_0) \hat{P}(y_0, y_1) ... \hat{P}(y_{N-1}, y_N)$$

To show that π is invariant for $\hat{P}$:

$$\sum_x \pi(x) \hat{P}(x,y) = \sum_x \pi(x) * \frac{\pi(y)}{\pi(x)} P(y,x) = \sum_x \pi(y) P(y,x) = \pi(y)$$

To prove $\hat{P}$ is irreducible, let x, y be two states. Since there exists a sequence of states with $P(x,x_1) P(x_1,x_2) ... P(x_{k-1},y) > 0$ by irreducibility of P we have that

$$\hat{P}(y,x_{k-1}) ... \hat{P}(x_2,x_1) \hat{P}(x_1,x) = \frac{\pi(x)}{\pi(y)} P(x,x_1) P(x_1,x_2) ... P(x_{k-1},y)$$

which means that $\hat{P}(y,x_{k-1}) ... \hat{P}(x_2,x_1) \hat{P}(x_1,x) > 0$ so $\hat{P}$ is irreducible.

□

Definition. If X has an invariant distribution, and we have the equality $\pi(x) = P(x,y) = \pi(y) P(y,x)$ for all x, y, then X is said to be *time reversible*. These are called the *detailed balance equations*. Equivalently, this says that $(X_0, X_1, ..., X_n)$ and $(X_n, X_{n-1}, ..., X_0)$ have the same distribution if we start at π (which is important as they certainly wouldn't if we start with, say, a constant distribution).

Intuitively, this says that if we watch a video of a Markov chain starting from π there would be no way to tell if the video is being played backwards or forwards.

Proposition. Suppose that μ is a distribution that satisfies $\mu(x) P(x,y) = \mu(y) P(y,x)$ for all x, y, then μ is an invariant distribution for P.

Proof. Because P is a stochastic matrix we compute

$$\sum_x \mu(x) P(x,y) = \sum_x \mu(y) P(y,x) = \mu(y)$$

□

Corollary. If some distribution satisfies $\mu(x) P(x,y) = \mu(y) P(y,x)$ then P is time reversible.

Example. Consider a random walk on the cycle $\mathbb{Z}_n$ with transition probabilities $P(i, (i+1) \bmod n) = \frac{2}{3}$ and $P(i, (i-1) \bmod n) = \frac{1}{3}$. Then $\pi(i) = \frac{1}{n}$ for all i is an invariant distirbution, but the chain is not reversible because we will see it drift to the right or left depending on which direction we play it in. I'll leave it to the reader to check that the detailed balance equations are not satisfied in this case.

Example. However, if the random walk is on $\{0, 1, ..., n-1\}$ with the same probabilities but with impossibility of wrapping around to the other side, then of course we will generally spend more time near the right, but this happens whether we go forwards or backwards in time, so it is actually reversible. The invariant measure that can satisfy the balance equations is $\lambda_i = 2^i$ times the appropriate constant that makes all of them sum to 1.

Definition. A graph is a collection of nodes connected by edges, it is connected if any 2 nodes are connected by a sequence of edges. We will do plenty of graph theory in the next sublevel, for now we just need this definition. Given a node x, $d(x)$ is the number of nodes connected to it.

Example. Consider a simple random walk on a graph such that from any vertex you have an equal chance to get to each neighbour vertex. Since the graph is connected, this implies P is irreducible. By solving the detailed balance equations to try to look for an invariant distribution. By noting that the balance equations give $\frac{\pi(x)}{d(x)} = \frac{\pi(y)}{d(y)}$, we note that setting a measure $d(x)$ for each x then normalizing gives such an invariant distribution.

5.3 Convergence to equilibrium

We already proved the result that says if X is a Markov chain on a finite state space I and $P(X_n = i)$ converges to some π_i as $n \to \infty$ for all $i \in I$ then π is an invariant distribution for X. We want to investigate when it is the case that P^n converges to the invariant distribution.

Note that, for example, if a Markov chain always jumps between 2 states with probability 1, then at odd times it will never be at the first state. This will clearly prevent convergence from happening. This motivates the following definition:

Definition. A state i in a Markov chain is said to be aperiodic if $\gcd\{n : P^n(i,i)\} = 1$.

Proposition. Let P be an irreducible transition matrix and let i be a state. Then the following are equivalent:

1. $\gcd\{n : P^n(i,i)\} = 1$

2. $P^n(i,i) > 0$ for all n large enough

3. $\gcd\{n : P^n(j,j)\} = 1$ for *every* state j and $P^n(i,i) > 0$

Proof. $1 \implies 2$: Set $D(i) = \{n \geq 1 : P^n(i,i) > 0\}$. We first show that $D(i)$ contains two consecutive integers. Note that if $m, n \in D(i)$ then so is $m+n$. Suppose the minimum distance between any 2 elements is $r \geq 2$ and $n, m \in D(i)$ with $n = m + r$. Suppose that $k \in D(i)$ with $k = lr + s$ and $l \in \mathbb{N}$ and $0 < s < r$, which exists since otherwise $D(i)$ would contain only multiples of r contradicting (1). Letting $a = (l+1)n$ and $b = (l+1)m + k$, we see that both a and b are contained in $D(i)$ and $a - b = r - s \in (0, r)$ contradicting minimality. since $D(i)$ therefore must contain two consecutive integers, $n_1, n_1 + 1$, it follows that $an_1 + b(n_1 + 1) \in D(i)$ for all a, b so every $n \geq n_1^2$ is also in $D(i)$.

$2 \implies 3$: If j is any other state distinct from i, then by irreducibility, i, j communicate so there exists m, n with $P^n(j,i) > 0$ and $P^m(i,j) > 0$. Then for all $s > 0$ sufficiently large we have

$$P^{s+n+m}(j,j) \geq P^n(j,i) P^s(i,i) P^m(i,j) > 0$$

which shows that $P^l(j,j) > 0$ for all l sufficiently large, and because the implication $2 \implies 1$ holds for every state (we didn't do this explicitly, but it's immediate from the definition) this means that $\gcd\{n : P^n(j,j)\} = 1$.

$3 \implies 1$: Obvious

□

Theorem. Suppose that P is irreducible and aperiodic and has invariant distribution π. Let $X \sim \text{Markov}(\lambda, P)$ where λ is any distribution. Then for all y we have

$$\lim_{n\to\infty} P(X_n = y) = \pi(y)$$

in particular, taking $\lambda = \delta_x$ we have

$$\lim_{n\to\infty} P^n(x,y) = \pi(y)$$

To motivate the strategy of the proof, consider the following card trick:

Shuffle 2 decks of cards (as 1 pile). Then reveal a card. Assign to each card a number: For 2-9 use its own number, 1 for aces, and 5 for the rest. Start with any number from 1 to 10 of your choice, and when each subsequent card is revealed count down by 1. If you get to 1, then look at the card and remember its corresponding number and when the next card is revealled say that number and start counting down again. To make this trick work, ask a friend to play along with a different starting number. Then when you get near the end of the deck, start saying the numbers out loud, and likely

they will be the same numbers as your friend despite you having picked a different starting number. It turns out the theorem has a very high chance to work - about 97%. The way it works is that if you agree at *any* point, you will agree permanently going forward.

Proof. Let $Y = (Y_n) \sim \text{Markov}(\pi, P)$ be independent of X and consider the process (X_n, Y_n) which is a markov chain on $I \times I$ with initial distribution $\lambda \times \pi$ and transition matrix $\tilde{P}$ given by

$$\tilde{P}((x_1, y_1), (x_2, y_2)) = P(x_1, x_2) P(y_1, y_2)$$

Then $\tilde{P}$ is irreducible because if (x, y) and (x', y') are two states in $I \times I$ then by irreducibility of P we have that there exists m and l such $P^m(x, x') > 0$ and $P^l(y, y') > 0$. By aperiodicity of P for all n large enough we have that

$$P^n(x, x') \geq P^m(x, x') P^{n-m}(x', x') > 0$$

and

$$P^n(y, y') \geq P^l(y, y') P^{n-m}(y', y') > 0$$

Therefore $\tilde{P}$ is irreducible. For $a \in I$ define

$$T := \inf\{n \geq 1 : (X_n, Y_n) = (a, a)\}$$

Then T is a stopping time for Markov chain (X, Y). Since $\tilde{\pi}(x, y) := \pi(x) \pi(y)$ is invariant for $\tilde{P}$, it follows from an earlier theorem that $\tilde{P}$ is positive recurrent which means (a, a) is recurrent. Therefore $P(T < \infty) = 1$.

Define a new process Z_n by X_n if $n < t$ and Y_n if $n \geq T$. This new process is Markov with initial distribution λ and transition matrix P by common sense. If $y \in I$, then since X and Z have the same distribution and Y is a stationary chain, we have

$$\begin{aligned} |P(X_n = y) - \pi(y)| &= |P(Z_n = y) - P(Y_n = y)| \\ &= |P(Z_n = y, T > n) + P(Z_n = y, T \leq n) - P(Y_n = y, T > n) - P(Y_n = y, T \leq n)| \\ &= |P(Z_n = y, T > n) - P(Y_n = y, T > n)| \leq P(T > n) \end{aligned}$$

Since $P(T > n) \to 0$ as $n \to \infty$ we are done.

□

Theorem. Suppose that P is an irreducible, null-recurrent and aperiodic transition matrix. Then for all x, y we have that $P^n(x, y) \to 0$ as $n \to \infty$.

Proof. Let Y be an independent copy of X and define $\tilde{P}$ analogously to the theorem above. As we showed already, $\tilde{P}$ is irreducible because P is irreducible and aperiodic. If $\tilde{P}$ is transient then the the statement follows because for every (x, y) we would have

$$\sum_n \tilde{P}^n((x, x), (y, y)) = \sum_n (P^n(x, y))^2 < \infty$$

which implies that $P^n(x, y) \to 0$ as $n \to \infty$.

Now suppose $\tilde{P}$ is recurrent. For $y \in I$ consider the measure ν_y defined by $E_y\left[\sum_{i=0}^{T_y - 1} 1(X_i = x)\right]$. Then since P is irreducible and recurrent it follows that ν_y is invariant for P by earlier work. Since P is null recurrent it means that $\nu_y(I) = E_y(T_y) = \infty$. Fix $M > 0$. Then there exists a finite set A such that $\nu_y(A) > M$ because $\nu_y(I) = \infty$. Now define a measure as $\mu(x) = \frac{\nu_y(x)}{\nu_y(A)} 1(x \in A)$. Then μ is a probability measure, and by the invariance of ν_y, we have that $\nu_y = \nu_y P^n$ for all n so we have

$$\mu P^n(z) = \sum_x \mu(x) P^n(x, z) \leq \sum_x \frac{\nu_y(x)}{\nu_y(A)} P^n(x, z) = \frac{\nu_y(z)}{\nu_y(A)}$$

Now let (X, Y) be Markov starting at $\mu \times \delta_x$ with transition matrix $\tilde{P}$. Let $T = \inf\{n \geq 1 : (X_n, Y_n) = (x, x)\}$. Then T is a finite stopping time with probability 1 because $\tilde{P}$ is recurrent. Defining $Z_n = \begin{cases} X_n & n < T \\ Y_n & n \geq T \end{cases}$ we see that Z is a Markov chain with initial distribution μ and transition matrix P similar to the theorem above. Therefore we obtain

$$P(Z_n = y) = \mu P^n(y) \leq \frac{\nu_y(y)}{\nu_y(A)} = \frac{1}{\nu_y(A)} < \frac{1}{M}$$

Then the result follows since

$$P_x(Y_n = y) \leq P(Z_n = y) + P(T > n)$$

Letting $n \to \infty$ and then letting $M \to \infty$ we have $\lim_{n\to\infty} P_x(Y_n = y) = 0$, so the result follows.

□

5.4 The ergodic theorem

Theorem. Let $X \sim \text{Markov}(\lambda, P)$ where P is an irreducible and positive recurrent matrix with invariant distribution π. Then for all x we have almost surely

$$\frac{1}{n}\sum_{i=0}^{n-1} 1(X_i = x) \to \pi(x)$$

as $n \to \infty$.

In the proof, we will use the fact that the strong law of large numbers from the probability course holds as long as $E[|X|]$ is finite, the proof of which is deferred to probability and measure/level 11.1.

Proof. Define $V_n(x) := \sum_{i=0}^{n-1} 1(X_i = x)$ for the total number of visits to x up to time $n-1$.

Since P is recurrent it follows that $T_x < \infty$ with probability 1 and by the strong Markov property the process X_{T_x+n} is $\text{Markov}(\delta_x, P)$ independent of $(X_0, ..., X_T)$. Since $T_x < \infty$ with probability 1 we have that the limit $\lim \frac{V_n(x)}{n}$ is almost surely the same when the initial distribution is λ and δ_x so assume it is δ_x.

Define $T_x^{(k+1)}$ as the time of the $k+1$'th visit to x after starting at δ_x, and $S_x^{(k)} := T_x^{(k)} - T_x^{(k-1)}$ (or 0 if the difference is infinite or undefined which happens with probability 0 unless $k = 0$). By definition of the return times we have $T_x^{(V_n(x)-1)} \leq n-1$ which means $S_x^{(1)} + ... + S_x^{(V_n(x)-1)} \leq n-1$. Similarly, $T_x^{(V_n(x))} \geq n$ and $S_x^{(1)} + ... + S_x^{(V_n(x))} \geq n$. By the strong Markov property, the lengths S_x are iid with expectation given by $E_x[T_x] = \frac{1}{\pi(x)}$. Then, by the strong law of large numbers, the mean of the first k S_x converges almost surely to $\frac{1}{\pi(x)}$. Now have

$$S_x^{(1)} + ... + S_x^{(V_n(x))} \geq n \geq S_x^{(1)} + ... + S_x^{(V_n(x)-1)}$$

Dividing through gives

$$\frac{S_x^{(1)} + ... + S_x^{(V_n(x))}}{V_n(x)} \geq \frac{n}{V_n(x)} \geq \frac{S_x^{(1)} + ... + S_x^{(V_n(x)-1)}}{V_n(x)}$$

Letting $n \to \infty$ we conclude that $\frac{V_n}{n} \to \pi(x)$ with probability 1 as required.

□